Ex:1	Private Cloud using OpenStack
24/6/2026	

AIM

To install and configure OpenStack using DevStack on an Ubuntu virtual machine, verify the OpenStack services through the command line, create a test virtual machine using the CirrOS image, and access the OpenStack Horizon web dashboard.

OBJECTIVES

- Install the required packages and Git.
- Create and configure the OpenStack stack user.
- Download/configure DevStack and prepare local.conf.
- Verify hardware virtualization/KVM support.
- Run the DevStack installation successfully.
- Verify OpenStack services, compute services, agents, images, flavors and networks.
- Create and verify an OpenStack instance named test-vm.
- Access OpenStack Horizon through a web browser.

REQUIREMENTS

- Ubuntu 26.04 LTS virtual machine
- Oracle VM VirtualBox
- Internet connectivity
- Minimum 4 vCPUs recommended for the lab setup
- Sufficient RAM and disk space for DevStack
- Git

THEORY

OpenStack is an open-source cloud computing platform used to provide infrastructure services such as compute, networking, storage and image management. DevStack is a set of scripts used to install and configure a development-oriented OpenStack environment.

The major services used in this experiment include Keystone for identity, Nova for compute, Neutron for networking, Glance for image management and Cinder for block storage. Horizon provides a browser-based graphical interface for managing these services.

PROCEDURE

Step 1 – Verify Git installation

```
git --version
```

Step 2 – Create the stack user

```
sudo useradd -s /bin/bash -d /opt/stack -m stack
echo "stack ALL=(ALL) NOPASSWD: ALL" | sudo tee /etc/sudoers.d/stack
sudo su - stack
id -un
```

Step 3 – Obtain DevStack

```
git clone https://opendev.org/openstack/devstack
cd ~/devstack
```

Step 4 – Configure DevStack

```
nano local.conf
cat local.conf
```

Step 5 – Verify virtualization support

```
egrep -c '(vmx|svm)' /proc/cpuinfo
lsmod | grep kvm
```

Step 6 – Run DevStack

```
cd ~/devstack
./stack.sh
```

Step 7 – Load OpenStack credentials

```
source openrc admin
```

Step 8 – Verify OpenStack services

```
openstack service list
openstack compute service list
openstack network agent list
```

Step 9 – Verify images, flavors and networks

```
openstack image list
openstack flavor list
openstack network list
```

Step 10 – Verify/create the test instance

```
openstack server list
openstack server show test-vm
```

Step 11 – Access the Horizon dashboard

```
Open Firefox/Chrome inside the Ubuntu VM.
Browse to: http://127.0.0.1/dashboard/
Log in as admin using the configured password.
```

DevStack Configuration Used

```
[[local|localrc]]
HOST_IP=10.0.2.15
GIT_BASE=https://opendev.org
ADMIN_PASSWORD=secret
DATABASE_PASSWORD=$ADMIN_PASSWORD
RABBIT_PASSWORD=$ADMIN_PASSWORD
SERVICE_PASSWORD=$ADMIN_PASSWORD
```

The above configuration was used during the successful DevStack deployment. The Horizon dashboard was later accessed locally through 127.0.0.1.

OBSERVATION

Verification	Observed Result	Status
Git	git version 2.53.0	Verified
KVM	kvm_intel, kvm and irqbypass modules present	Verified
DevStack	stack.sh completed successfully	Successful
OpenStack services	Keystone, Placement, Glance, Nova, Neutron, Cinder and legacy Nova listed	Verified
Compute services	Nova scheduler/conductor/compute services shown as enabled and up	Verified
Image	cirros-0.6.3-x86_64-disk – active	Verified
Instance	test-vm – ACTIVE	Verified
Instance network	private=10.0.0.55	Verified
Horizon	OpenStack dashboard accessible at http://127.0.0.1/dashboard/	Verified

OUTPUT

The following screenshots are included as practical evidence of the installation, configuration, verification and successful OpenStack operation.

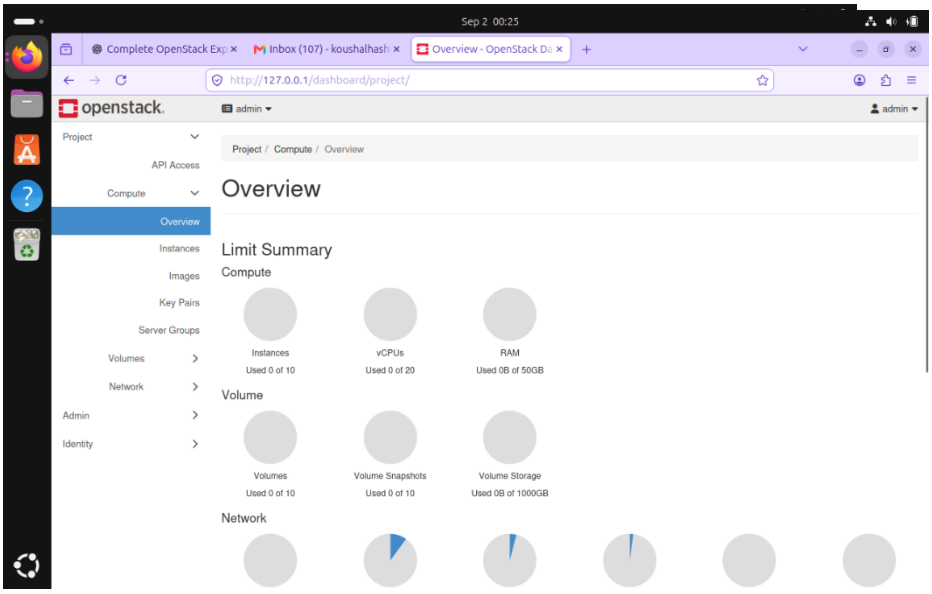


Figure : OpenStack Horizon web dashboard successfully opened in Firefox.

CONCLUSION

Thus, the basic installation, configuration and operation of an OpenStack cloud environment using DevStack were successfully completed.